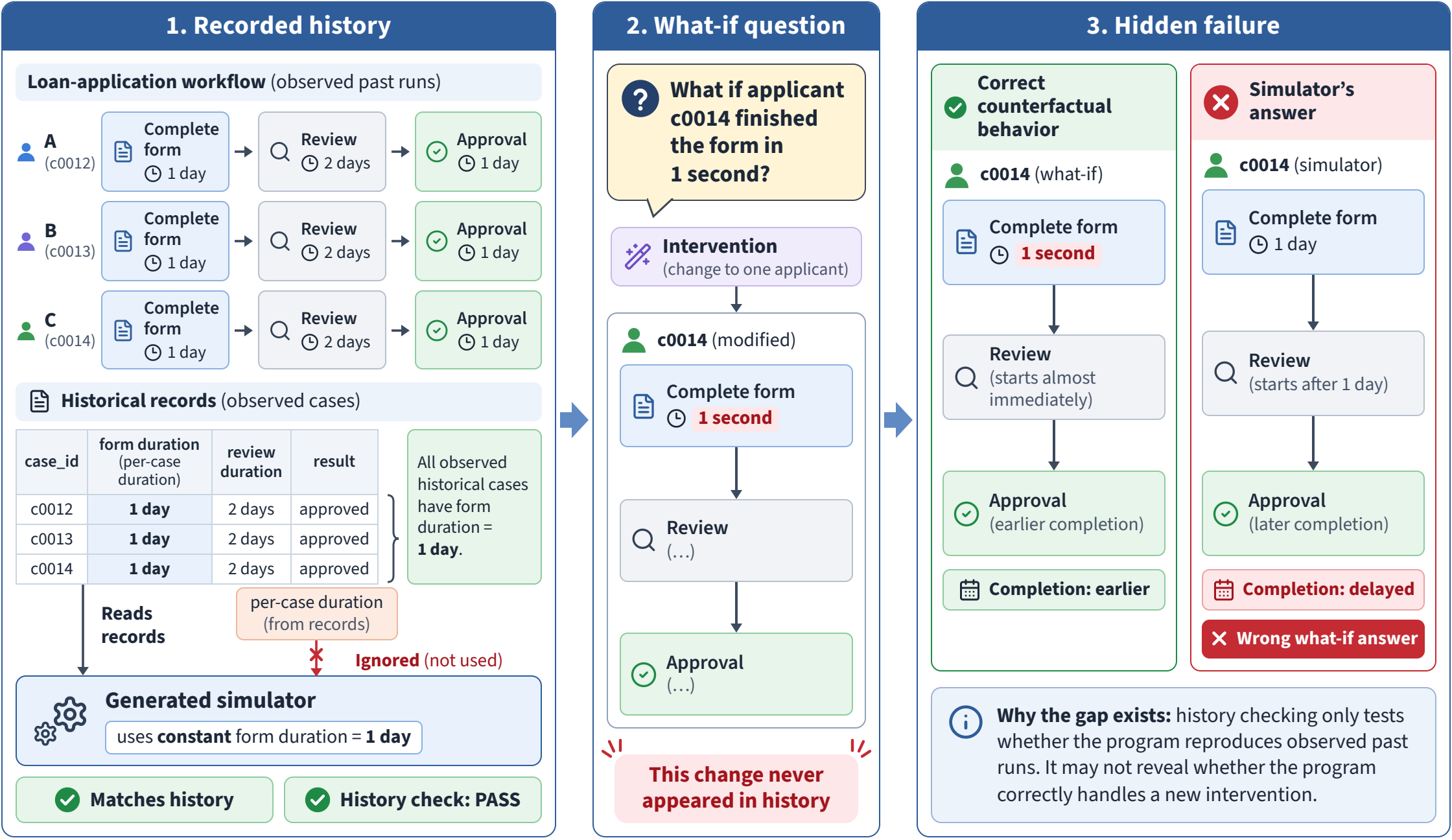

1. Recorded history
Loan-application workflow (observed past runs)
A (c0012)
Complete form 1 day
Review 2 days
Approval 1 day
B (c0013)
Complete form 1 day
Review 2 days
Approval 1 day
C (c0014)
Complete form 1 day
Review 2 days
Approval 1 day
Historical records (observed cases)
case_id
form duration (per-case duration)
review duration
result
c0012
1 day
2 days
approved
c0013
1 day
2 days
approved
c0014
1 day
2 days
approved
All observed historical cases have form duration = 1 day.
Reads records
per-case duration (from records)
Ignored (not used)
Generated simulator
uses constant form duration = 1 day
Matches history
History check: PASS
2. What-if question
What if applicant c0014 finished the form in 1 second?
Intervention (change to one applicant)
c0014 (modified)
Complete form 1 second
Review (…)
Approval (…)
This change never appeared in history
3. Hidden failure
Correct counterfactual behavior
c0014 (what-if)
Complete form 1 second
Review (starts almost immediately)
Approval (earlier completion)
Completion: earlier
Simulator's answer
c0014 (simulator)
Complete form 1 day
Review (starts after 1 day)
Approval (later completion)
Completion: delayed
Wrong what-if answer
Why the gap exists: history checking only tests whether the program reproduces observed past runs. It may not reveal whether the program correctly handles a new intervention.